KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3341 | 3 Credits: 4 | Program Core | Source-grounded

# Discrete Mathematics

 Students will be able to develop logical thinking for solving computational problems.

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3341 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 3341.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

## Official course information

**Course information**

| Item | Official information |
| --- | --- |
| Programme | Diploma in Artificial Intelligence & Machine Learning |
| Course code | 3341 |
| Course title | Discrete Mathematics |
| Semester | 3 Credits: 4 |
| Course category | Program Core |
| Periods per week | 4 (L:4 T:0 P:0) Periods per semester: 60 |

| Item | Official information |
| --- | --- |
| Periods per semester | 60 |
| Credits | 4 |
| Prerequisite | See the official syllabus PDF |
| Assessment | Use the official assessment section reproduced below |

### Modules

4 official module sections detected.

### Topic entries

21 source-aligned topic entries retained.

### Study mode

Theory and application emphasis.

STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

# Objectives, outcomes and mapping

## Course objectives

-  Students will be able to develop logical thinking for solving computational problems.

**Course outcomes**

| CO | Official outcome | Study level |
|---|---|---|
| CO1 | 16 Applying solve problems Applying | See official syllabus |
| CO2 | Demonstrate sets, relations and functions 16 Solve counting problems by applying elementary | See official syllabus |
| CO3 | 16 Applying counting techniques | See official syllabus |
| CO4 | Summarize the abstract algebraic systems 10 Understanding | See official syllabus |

**CO–PO mapping evidence**

| Row | Extracted official mapping |
|---|---|
| CO1 | CO1 3 |
| CO2 | CO2 3 |
| CO3 | CO3 3 |
| CO4 | CO4 3 |

COVERAGE AUDIT

# Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

**Module coverage**

| Module | Official heading | Topic entries | Hours |
|---|---|---|---|
| Module 1 | Apply the basic concepts of mathematical logic to solve problems | 5 | As specified |
| Module 2 | Demonstrate sets, relations and functions | 7 | As specified |
| Module 3 | Solve counting problems by applying elementary counting techniques | 6 | As specified |
| Module 4 | Summarize the abstract algebraic systems | 3 | As specified |

MODULE 1

# Apply the basic concepts of mathematical logic to solve problems

This module is presented from the official syllabus scope for Discrete Mathematics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Apply the basic concepts of mathematical logic to solve problems
- Official duration: As specified
- Official topic entries captured: 5

- The full source excerpt is preserved below for coverage checking.

## Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical

Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, describe the concept of propositional logic 2 understanding solve problems using truth tables and logical should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## 3 Applying operations Prove logical equivalence, contingency,

3 Applying operations Prove logical equivalence, contingency, is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Applying operations Prove logical equivalence, contingency, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 applying operations prove logical equivalence, contingency, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Applying operations Prove logical equivalence, contingency, means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ 3 Applying operations Prove logical equivalence, contingency, □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## 4 Applying tautology and contradictions

4 Applying tautology and contradictions is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Applying tautology and contradictions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 applying tautology and contradictions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 4 Applying tautology and contradictions means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-[illegible] 4 Applying tautology and contradictions [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Describe the concept of predicate Logic

Describe the concept of predicate Logic is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Describe the concept of predicate Logic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, describe the concept of predicate logic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Describe the concept of predicate Logic means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-[illegible] Describe the concept of predicate Logic [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle.**

Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. is an official topic in Module 1 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, apply inference rules to solve problems 4 applying mathematical logic: propositional logic, connectives and truth table, implication, tautology, contradiction, contingency, equivalence - inverse, converse, contrapositive, duality principle, disjunction and conjunction, predicate logic, well formed formula, quantifiers-universal & existential, natural deduction, rules of inference, methods of proofs, resolution principle. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Apply inference rules to solve problems 4 Applying Mathematical logic: Propositional logic, Connectives and truth table, Implication, Tautology, Contradiction, Contingency, Equivalence - inverse, converse, contrapositive, Duality principle, Disjunction and Conjunction, Predicate Logic, Well formed formula, Quantifiers-universal & existential, Natural deduction, Rules of Inference, Methods of proofs, Resolution principle. □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**Official syllabus detail and terminology**

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Apply the basic concepts of mathematical logic to solve problems

  M1.01      Describe the concept of propositional Logic          2
Understanding
             Solve problems using truth tables and logical
  M1.02                                                           3
Applying
             operations
             Prove logical equivalence, contingency,
  M1.03                                                           4
Applying
             tautology and contradictions
  M1.04      Describe the concept of predicate Logic              3
Understanding

  M1.05      Apply inference rules to solve problems              4
Applying
Contents:
Mathematical logic:
Propositional logic, Connectives and truth table, Implication, Tautology,
Contradiction,
Contingency, Equivalence - inverse, converse, contrapositive, Duality principle,
Disjunction
and Conjunction,
Predicate Logic, Well formed formula, Quantifiers-universal & existential,
Natural
deduction, Rules of Inference, Methods of proofs, Resolution principle.
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

# Demonstrate sets, relations and functions

This module is presented from the official syllabus scope for Discrete Mathematics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Demonstrate sets, relations and functions
- Official duration: As specified
- Official topic entries captured: 7
- The full source excerpt is preserved below for coverage checking.

## Outline the basic principles of set

Outline the basic principles of set is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Outline the basic principles of set using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, outline the basic principles of set should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Outline the basic principles of set means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Outline the basic principles of set □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## Demonstrate the operations on sets

Demonstrate the operations on sets is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate the operations on sets using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate the operations on sets should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Demonstrate the operations on sets means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] Demonstrate the operations on sets [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Apply inclusion exclusion principle

Apply inclusion exclusion principle is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Apply inclusion exclusion principle using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, apply inclusion exclusion principle should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Apply inclusion exclusion principle means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Apply inclusion exclusion principle □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## Demonstrate the concept of relations

Demonstrate the concept of relations is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate the concept of relations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate the concept of relations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Demonstrate the concept of relations means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Demonstrate the concept of relations □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## Identify the equivalence relations 2 Applying Illustrate the equivalence classes from

Identify the equivalence relations 2 Applying Illustrate the equivalence classes from is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Identify the equivalence relations 2 Applying Illustrate the equivalence classes from using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, identify the equivalence relations 2 applying illustrate the equivalence classes from should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Identify the equivalence relations 2 Applying Illustrate the equivalence classes from means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Identify the equivalence relations 2 Applying Illustrate the equivalence classes from □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

## 3 Applying equivalence relation

3 Applying equivalence relation is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Applying equivalence relation using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 applying equivalence relation should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Applying equivalence relation means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-[illegible] 3 Applying equivalence relation [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions**

Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions is an official topic in Module 2 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, demonstrate the concept of functions 3 applying set theory:-set representation, types of sets, set operations, algebra of sets, inclusion- exclusion principle relations: representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class functions: mappings, injection and surjections, composition of functions, inverse functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ Demonstrate the concept of functions 3 Applying Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets, Inclusion- exclusion principle Relations: Representation - matrix -graph, types of relations, properties, equivalence relations, equivalence class Functions: Mappings, Injection and Surjections, Composition of functions, Inverse functions □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Demonstrate sets, relations and functions

  M2.01      Outline the basic principles of set                  2
Understanding
  M2.02      Demonstrate the operations on sets                  2
Applying
  M2.03      Apply inclusion exclusion principle                 2
Applying
  M2.04      Demonstrate the concept of relations                2
Understanding
  M2.05      Identify the equivalence relations                  2
Applying
             Illustrate the equivalence classes from
  M2.06                                                          3
Applying
             equivalence relation
  M2.07      Demonstrate the concept of functions                3
Applying
             Series Test – I                                     1
Contents:
Set theory:-Set representation, Types of Sets, Set operations, Algebra of sets,
Inclusion-
exclusion principle
Relations: Representation - matrix -graph, types of relations, properties,
equivalence
relations, equivalence class
Functions: Mappings, Injection and Surjections, Composition of functions, Inverse
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# Solve counting problems by applying elementary counting techniques

This module is presented from the official syllabus scope for Discrete Mathematics. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Solve counting problems by applying elementary counting techniques
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

## Applying techniques using the product and sum rules

Applying techniques using the product and sum rules is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Applying techniques using the product and sum rules using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, applying techniques using the product and sum rules should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What Applying techniques using the product and sum rules means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Applying techniques using the product and sum rules □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

## Applying permutations & combinations

Applying permutations & combinations is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Applying permutations & combinations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, applying permutations & combinations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Applying permutations & combinations means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] Applying permutations & combinations [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Applying pigeon-hole principle 2

Applying pigeon-hole principle 2 is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Applying pigeon-hole principle 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, applying pigeon-hole principle 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Applying pigeon-hole principle 2 means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] Applying pigeon-hole principle 2 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

## Explain Binomial theorem Understanding Make use of recurrence relations to solve 3

Explain Binomial theorem Understanding Make use of recurrence relations to solve 3 is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Explain Binomial theorem Understanding Make use of recurrence relations to solve 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, explain binomial theorem understanding make use of recurrence relations to solve 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Explain Binomial theorem Understanding Make use of recurrence relations to solve 3 means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Explain Binomial theorem Understanding Make use of recurrence relations to solve 3 □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## Applying problems 2

Applying problems 2 is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Applying problems 2 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, applying problems 2 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Applying problems 2 means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] Applying problems 2 [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions**

Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions is an official topic in Module 3 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, illustrate generating functions applying basic counting principle, rule of sum, rule of product permutations, combinations. pigeon hole principle , binomial theorem (without proof). recurrence relation, generating functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions means in the official course context |

| Aspect | What to prepare |
|---|---|
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Illustrate generating functions Applying Basic counting principle, Rule of Sum, Rule of Product Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without proof). Recurrence relation, Generating functions □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Solve counting problems by applying elementary counting techniques


                                                                3
          Solve counting problems by applying the
 M3.01                                                                  Applying
          techniques using the product and sum rules

          Solve counting problems by applying                   3
 M3.02                                                                  Applying
          permutations & combinations
          Solve counting problems by applying the               3
 M3.03                                                                  Applying
          pigeon-hole principle
                                                                2
 M3.04    Explain Binomial theorem
Understanding

          Make use of recurrence relations to solve             3
 M3.05                                                                  Applying
          problems
                                                                2
 M3.06    Illustrate generating functions                               Applying
Contents:
Basic counting principle, Rule of Sum, Rule of Product
Permutations, Combinations. Pigeon hole Principle , Binomial Theorem (without
proof).
Recurrence relation, Generating functions
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

# Summarize the abstract algebraic systems

This module is presented from the official syllabus scope for Discrete Mathematics. Use the topic cards for learning and the source transcript for exact coverage.

| **As specified** | **CO-linked** |
|---|---|
| official hours | see outcomes |

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Summarize the abstract algebraic systems
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## Understanding isomorphism Describe the abstract algebraic systems - 5

Understanding isomorphism Describe the abstract algebraic systems - 5 is an official topic in Module 4 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Understanding isomorphism Describe the abstract algebraic systems - 5 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, understanding isomorphism describe the abstract algebraic systems - 5 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Understanding isomorphism Describe the abstract algebraic systems - 5 means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Understanding isomorphism Describe the abstract algebraic systems - 5 □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

**Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3**

Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3 is an official topic in Module 4 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

## Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

## Study points

- Define or identify Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, groups, subgroups, semigroups, monoids & understanding abelian groups outline the concept of lattice by definition, 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3 means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3 □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

**Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 https://archive.nptel.ac.in/courses/111/106/111106086/**

Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 https://archive.nptel.ac.in/courses/111/106/111106086/ is an official topic in Module 4 of Discrete Mathematics. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 https://archive.nptel.ac.in/courses/111/106/111106086/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, understanding properties algebraic system - properties, homomorphism and isomorphism, group, semigroup, subgroup, monoid, abelian group (simple theorems) lattice -definition, properties, types of lattice text / reference: t/r book title / author t1 discrete and combinatorial mathematics(an applied introduction),ralph p grimaldi, b v ramana, 5th edition, pearson t2 kenneth h.rosen,discrete mathematics and its application with combinatorics and graph theory,seventh edition,mgh,2011 t3 “discrete mathematics with applications”,thomas koshy,academic press,2005 online resources: sl. no. website link 1 https://archive.nptel.ac.in/courses/111/106/111106086/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 https://archive.nptel.ac.in/courses/111/106/111106086/ means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-￼￼ Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 https://archive.nptel.ac.in/courses/111/106/111106086/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Summarize the abstract algebraic systems
             Differentiate homomorphism and                      2
  M4.01
Understanding
             isomorphism
               Describe the abstract algebraic systems -             5
   M4.02       Groups, subgroups, Semigroups, Monoids &
Understanding
               abelian groups
               Outline the concept of lattice by definition,         3
   M4.03
Understanding
               properties
               Series test II                                        1

Contents:
Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup,
subgroup, monoid, abelian group (simple theorems)

Lattice -Definition, properties, Types of lattice

Text / Reference:

  T/R                                    Book Title / Author

  T1       Discrete and combinatorial mathematics(An applied introduction),Ralph
P
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

## Concept and formula bank

### Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

**Define or explain Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain 3 Applying operations Prove logical equivalence, contingency,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Applying operations Prove logical equivalence, contingency,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain 4 Applying tautology and contradictions. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Applying tautology and contradictions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain Describe the concept of predicate Logic. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Describe the concept of predicate Logic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

## Module 2

**Define or explain Outline the basic principles of set. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Outline the basic principles of set*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

**Define or explain Demonstrate the operations on sets. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Demonstrate the operations on sets*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

**Define or explain Apply inclusion exclusion principle. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Apply inclusion exclusion principle*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

**Define or explain Demonstrate the concept of relations. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Demonstrate the concept of relations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 3

**Define or explain Applying techniques using the product and sum rules. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Applying techniques using the product and sum rules*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Applying permutations & combinations. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Applying permutations & combinations*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Applying pigeon-hole principle 2. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Applying pigeon-hole principle 2*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain Explain Binomial theorem Understanding Make use of recurrence relations to solve 3. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Explain Binomial theorem Understanding Make use of recurrence relations to solve 3*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

## Module 4

**Define or explain Understanding isomorphism Describe the abstract algebraic systems - 5. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Understanding isomorphism Describe the abstract algebraic systems - 5*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

**Define or explain Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Groups, subgroups, Semigroups, Monoids & Understanding abelian groups Outline the concept of lattice by definition, 3*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 https://archive.nptel.ac.in/courses/111/106/111106086/. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Understanding properties Algebraic System - Properties, Homomorphism and Isomorphism, Group, semigroup, subgroup, monoid, abelian group (simple theorems) Lattice -Definition, properties, Types of lattice Text / Reference: T/R Book Title / Author T1 Discrete and combinatorial mathematics(An applied introduction),Ralph P Grimaldi, B V Ramana, 5th edition, Pearson T2 Kenneth H.Rosen,Discrete Mathematics and its application with combinatorics and graph theory,seventh edition,MGH,2011 T3 “Discrete Mathematics with applications”,Thomas Koshy,Academic press,2005 Online Resources: Sl. No. Website Link 1 https://archive.nptel.ac.in/courses/111/106/111106086/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

# Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **Describe the concept of propositional Logic 2 Understanding Solve problems using truth tables and logical**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **Outline the basic principles of set**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **Applying techniques using the product and sum rules**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 4

1-mark definition: State the meaning of **Understanding isomorphism Describe the abstract algebraic systems - 5**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

## Glossary

**Study glossary**

| Term | Meaning in this lesson |
|---|---|
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module | An official syllabus unit with defined topics and hours. |
| CIE | Continuous Internal Evaluation. |
| ESE | End Semester Examination. |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

SOURCE-SUPPORTED REFERENCES

# References and web resources

## References listed in the official PDF

References are included in the official source PDF.

## Web resources listed in the official PDF

- Sl. No. Website Link
- https://archive.nptel.ac.in/courses/111/106/111106086/

IN-PAGE SEARCH

# Search this lesson

Prepared from the supplied official SITTTR syllabus PDF for Course 3341. Verify institutional updates against the current official curriculum.